

Composite Design Pattern

Case Study 2

Spotify Playlist System Case Study

Why do we need a Composite?



Type Checking Nightmare

Handling items manually means creating endless 'if-else' checks to distinguish between individual songs and folders.



Deeply Nested Logic

Folders inside folders become messy and hard to manage. Simple recursion is impossible without a unified structure.

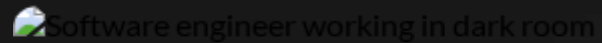


Maintenance Nightmare

Adding a single new audio type (like Podcasts) requires changing every iteration loop in the entire project.

| The Composite Pattern Solution

Uniform Treatment ☒



We stop using complex type-checks. Instead, we treat individual songs and playlist folders as exactly the same thing.

Recursive Playback ↻

Folders automatically delegate commands to their children. The client code is completely isolated from the hierarchical complexity!

Open/Closed ☒

No more manual iteration. We can add new playable items without ever changing the core playback engine.

Composite vs Flyweight

Feature	Composite Pattern 	Flyweight Pattern 
Focus	Structural Hierarchy (Tree Structure)	Memory Optimization (Object Sharing)
Playback Timing	Recursive execution on children	Immediate state retrieval
Use Case	Nested items with many levels	Millions of similar, small objects
Architecture	Part-Whole Relationship	Intrinsic vs Extrinsic State

System Architecture & UML

Component (Playable) 📁

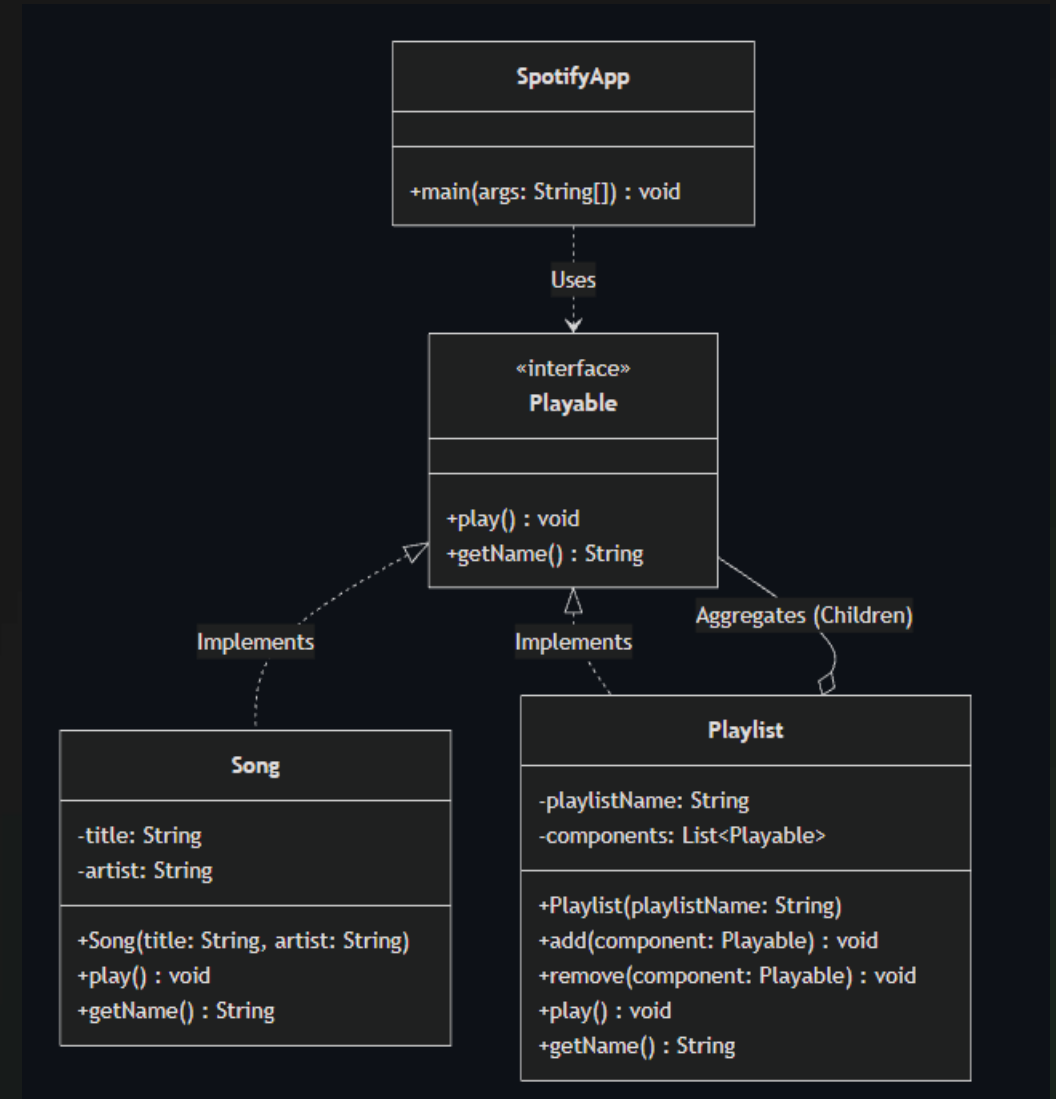
Standardizes the playback methods (e.g., play(), getName()) for all objects.

Leaf (Song) 🎵

Represents the individual tracks. It performs the actual playback logic.

Composite (Playlist) 📁

Maintains the collection of children. It orchestrates the recursive playback across the tree.



Java Implementation Details

1. Playable Interface

```
public interface Playable {  
    void play();  
    String getName();  
}
```

2. Composite Playlist

```
public void play() {  
    for (Playable p : components) {  
        p.play();  
    }  
}
```

Final Execution Output

```
> SYSTEM START: Playing the Master Playlist...  
--- Opening Playlist: My Favorites 2024 ---  
    --- Opening Playlist: Pop Hits ---  
        🎵 Playing Song: [Shape of You] by Ed Sheeran  
        🎵 Playing Song: [Blinding Lights] by The Weeknd  
    --- Opening Playlist: Classic Rock ---  
        🎵 Playing Song: [Hotel California] by Eagles
```

Recursive traversal completed successfully across all levels.

Thank You!

Composite Design Pattern Project

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